

Assignment 19: Graphing Quadratics

DO NOT WRITE ON THIS PAPER

Part A

Use the following quadratic equation to answer the questions below.

$$y = (x - 4)(x + 2)$$

- (a) What form is this equation in?
- (b) What information does this form of the equation give you (type and coordinate point(s))?
- (c) What is the vertex?
- (d) Does the graph open up or down?
- (e) How many solutions does this equation have?

Part B

Use the following quadratic equation to answer the questions below.

$$y = x^2 - 4x + 4$$

- (a) What form is this equation in?
- (b) What information does this form of the equation give you (type and coordinate point(s))?
- (c) What is the vertex?
- (d) Does the graph open up or down?
- (e) How many solutions does this equation have?

Part C

Use the following quadratic equation to answer the questions below.

$$y = -(x + 1)^2 - 2$$

- (a) What form is this equation in?
- (b) What information does this form of the equation give you (type and coordinate point(s))?
- (c) What is the vertex?
- (d) Does the graph open up or down?
- (e) How many solutions does this equation have?

Question 1

Big Question: What does the a-term tell us about a quadratic? What does it tell us about the vertex?

Suppose a company's revenue is given by a quadratic function of the number of widgets they create:

$$r(q) = -2q^2 + 40q + 100$$

How many widgets should the company make to maximize the revenue?

What is the maximum revenue?

Question 2 Work

Graph this quadratic equation.

$$y = (x - 2)^2 - 5$$

Question 2 Answer

Write your answer
in your Class Notes

Question 3 Work

Graph this quadratic equation.

$$y = -(x - 5)(x + 1)$$

Question 3 Answer

Write your answer
in your Class Notes